

CBSE 2027 Maths — Predicted Paper (Set 3 of 10)

Based on the highly weighted CBSE syllabus chapters such as Calculus, Algebra, and Vectors & 3D Geometry [1, 2], this predicted paper (Set 3) tests heavily repeated concepts using all-new, high-probability scenarios.

SECTION A: Objective Type Questions (1 Mark Each x 20 = 20 Marks)

Q1. If the function $f(x) = \frac{x-1}{x+1}$ is defined for $x \neq -1$, find the value of $f(f(x))$. * **Chapter:** Relations and Functions * **Probability Tag: Medium**

Q2. Write the principal value of $\sec^{-1}(-2)$. * **Chapter:** Inverse Trigonometric Functions * **Probability Tag: Very High**

Q3. If A is a square matrix of order 3 and $|3A| = k|A|$, what is the value of k ? * **Chapter:** Matrices and Determinants * **Probability Tag: Very High**

Q4. What is the value of the determinant $\begin{vmatrix} \sin^2 x & \cos^2 x & 1 \\ \cos^2 x & \sin^2 x & 1 \\ -10 & 12 & 2 \end{vmatrix}$? * **Chapter:** Determinants * **Probability Tag: High**

Q5. Find the derivative of $e^{\sin^{-1} x}$ with respect to x . * **Chapter:** Continuity and Differentiability * **Probability Tag: High**

Q6. Find the slope of the normal to the curve $y = 2x^2 + 3 \sin x$ at $x = 0$. * **Chapter:** Application of Derivatives * **Probability Tag: Medium**

Q7. Evaluate the definite integral: $\int_0^1 \frac{dx}{1+x^2}$. * **Chapter:** Integrals * **Probability Tag: Very High**

Q8. Find the area of the region bounded by the curve $y = \cos x$ between $x = 0$ and $x = \pi$ and the x-axis. * **Chapter:** Application of Integrals * **Probability Tag: High**

Q9. What is the order of the differential equation representing the family of all non-vertical straight lines in a plane? * **Chapter:** Differential Equations * **Probability Tag: Medium**

Q10. Find the position vector of the midpoint of the line segment joining the points $A(2, 3, 4)$ and $B(4, 1, -2)$. * **Chapter:** Vector Algebra * **Probability Tag: Very High**

Q11. Find a unit vector perpendicular to both the vectors $\hat{i} + \hat{j}$ and $\hat{j} + \hat{k}$. * **Chapter:** Vector Algebra * **Probability Tag: High**

Q12. Find the direction cosines of the line joining the points $(1, -1, 1)$ and $(2, 3, 1)$. * **Chapter:** Three-Dimensional Geometry * **Probability Tag: High**

Q13. Write the Cartesian equation of the x-axis in 3D space. * **Chapter:** Three-Dimensional Geometry * **Probability Tag: Medium**

Q14. In a Linear Programming Problem, the region represented by the non-negativity constraints $x \geq 0, y \geq 0$ restricts the feasible region to which quadrant? * **Chapter:** Linear Programming * **Probability Tag: Very High**

Q15. If $P(A) = 0.3$, $P(B) = 0.6$ and A and B are independent events, find $P(A \cup B)$. *

Chapter: Probability * **Probability Tag:** Very High

Q16. Evaluate: $\int \frac{\sec^2 x}{\sqrt{\tan x}} dx$. * **Chapter:** Integrals * **Probability Tag:** High

Q17. Check whether the relation R in set $A = \{1, 2, 3\}$ defined as $R = \{(1, 2), (2, 1)\}$ is transitive. * **Chapter:** Relations and Functions * **Probability Tag:** High

Q18. Find the integrating factor (I.F.) of the linear differential equation $x \frac{dy}{dx} + 2y = x^2$. *

Chapter: Differential Equations * **Probability Tag:** Very High

Q19. (Assertion-Reason): Assertion (A): If A and B are symmetric matrices of the same order, then $AB - BA$ is a skew-symmetric matrix. **Reason (R):** For any two matrices A and B , $(AB)^T = B^T A^T$. * **Chapter:** Matrices * **Probability Tag:** High

Q20. (Assertion-Reason): Assertion (A): The function $f(x) = e^x$ is strictly increasing on $\mathbb{R}$. **Reason (R):** A function $f(x)$ is strictly increasing on an interval if $f'(x) > 0$ for all x in that interval. * **Chapter:** Application of Derivatives * **Probability Tag:** High

SECTION B: Very Short Answer Type Questions (2 Marks Each x 5 = 10 Marks)

Q21. Prove that $2 \tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{7}\right) = \tan^{-1}\left(\frac{31}{17}\right)$. * **Chapter:** Inverse Trigonometric Functions * **Probability Tag:** High

Q22. The volume of a cube is increasing at a rate of $8 \text{ cm}^3/\text{s}$. How fast is the surface area increasing when the length of an edge is 12 cm ? * **Chapter:** Application of Derivatives * **Probability Tag:** Very High

Q23. If $\vec{a}$, $\vec{b}$, and $\vec{c}$ are mutually perpendicular unit vectors, find the exact magnitude of $\vec{a} + \vec{b} + \vec{c}$. * **Chapter:** Vector Algebra * **Probability Tag:** Very High

Q24. Evaluate the definite integral: $\int_0^{\pi/2} \log(\tan x) dx$. * **Chapter:** Integrals * **Probability Tag:** High

Q25. An urn contains 10 black and 5 white balls. Two balls are drawn from the urn one after the other without replacement. What is the probability that both drawn balls are black? * **Chapter:** Probability * **Probability Tag:** High

SECTION C: Short Answer Type Questions (3 Marks Each x 6 = 18 Marks)

Q26. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = 10x + 7$. Find the function $g : \mathbb{R} \rightarrow \mathbb{R}$ such that $gof = fog = I_{\mathbb{R}}$ (the identity function on $\mathbb{R}$). * **Chapter:** Relations and Functions * **Probability Tag:** High

Q27. If $A = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$, then find the value of α lying in the interval $[0, \pi/2]$ for which $A + A' = I$. * **Chapter:** Matrices * **Probability Tag:** Medium

Q28. If $x = a(\theta - \sin \theta)$ and $y = a(1 + \cos \theta)$, find the value of $\frac{dy}{dx}$ at $\theta = \frac{\pi}{3}$. * **Chapter:** Continuity and Differentiability * **Probability Tag: Very High**

Q29. Evaluate: $\int \frac{x \sin^{-1} x}{\sqrt{1-x^2}} dx$. * **Chapter:** Integrals * **Probability Tag: High**

Q30. Solve the differential equation: $x \log x \frac{dy}{dx} + y = \frac{2}{x} \log x$. * **Chapter:** Differential Equations * **Probability Tag: High**

Q31. Solve the following Linear Programming Problem graphically: Minimize $Z = -3x + 4y$ Subject to the constraints: $x + 2y \leq 8$ $3x + 2y \leq 12$ $x \geq 0, y \geq 0$ * **Chapter:** Linear Programming * **Probability Tag: Very High**

SECTION D: Long Answer Type Questions (5 Marks Each x 4 = 20 Marks)

Q32. Evaluate the following definite integral using properties of integration: $\int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx$. * **Chapter:** Integrals * **Probability Tag: Very High**

Q33. Find the coordinates of the foot of the perpendicular drawn from the point $P(0, 2, 3)$ to the line $\frac{x+3}{5} = \frac{y-1}{2} = \frac{z+4}{3}$. Also, find the length of the perpendicular segment. * **Chapter:** Three-Dimensional Geometry * **Probability Tag: Very High**

Q34. Using the method of integration, find the area of the region bounded by the triangle whose vertices are $A(-1, 0)$, $B(1, 3)$ and $C(3, 2)$. * **Chapter:** Application of Integrals * **Probability Tag: Very High**

Q35. Let $A = \mathbb{R} - \{3\}$ and $B = \mathbb{R} - \{1\}$. Consider the function $f: A \rightarrow B$ defined by $f(x) = \frac{x-2}{x-3}$. Show that f is both one-one and onto. * **Chapter:** Relations and Functions * **Probability Tag: High**

SECTION E: Case-Study Based Questions (4 Marks Each x 3 = 12 Marks)

Q36. Case Study 1: Matrices & Determinants (Cryptography) A school's cyber club uses matrix multiplication to encrypt secret passwords. They use a 2×2 encoding matrix A . To decode, they multiply by the inverse matrix A^{-1} . A letter sequence is translated into numbers and represented as a 2×1 matrix X . The encoding matrix used is $A = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}$.
 (i) Find the determinant of the encoding matrix A . (1 Mark) (ii) Find the decoding matrix A^{-1} . (1 Mark) (iii) If the club intercepts the encoded message matrix $Y = \begin{pmatrix} 5 \\ 8 \end{pmatrix}$, find the original message matrix X using the relation $AX = Y$. (2 Marks) * **Chapter:** Matrices and Determinants * **Probability Tag: Medium**

Q37. Case Study 2: Application of Derivatives (Architectural Design) An architect is designing a stadium window in the shape of a rectangle surmounted by a semi-circle. To maintain structural integrity, the total perimeter of the window is fixed at exactly **10 meters**. Let x be the width of the rectangular part (which also serves as the diameter of the semi-

circle) and y be the height of the rectangular part. (i) Express the perimeter equation in terms of x and y , and write y in terms of x . (1 Mark) (ii) Express the total area A of the window purely as a function of x . (1 Mark) (iii) Find the dimensions x and y that will maximize the area of the window to let in the maximum possible amount of light. (2 Marks) * **Chapter:**

Application of Derivatives * **Probability Tag: Very High**

Q38. Case Study 3: Probability (Total Probability & Bayes' Theorem) During a weekend trip, a group of friends uses a standard deck of 52 playing cards. While traveling, one card is accidentally lost from the well-shuffled pack. From the remaining 51 cards, two cards are drawn at random (without replacement) and it is discovered that both are diamonds. [3, 4] (i) What is the probability that the accidentally lost card was a diamond? (1 Mark) (ii) Let E_1 be the event that the lost card is a diamond, and E_2 be the event that the lost card is not a diamond. What is $P(E_2)$? (1 Mark) (iii) Given the evidence that the two drawn cards are diamonds, use Bayes' theorem to find the exact probability that the lost card was originally a diamond. (2 Marks) * **Chapter:** Probability * **Probability Tag: Very High**

CBSE Class 12 (2027) — AI-Predicted via PYQ/Blueprint Analysis. Probability-weighted guidance, not actual exam questions.